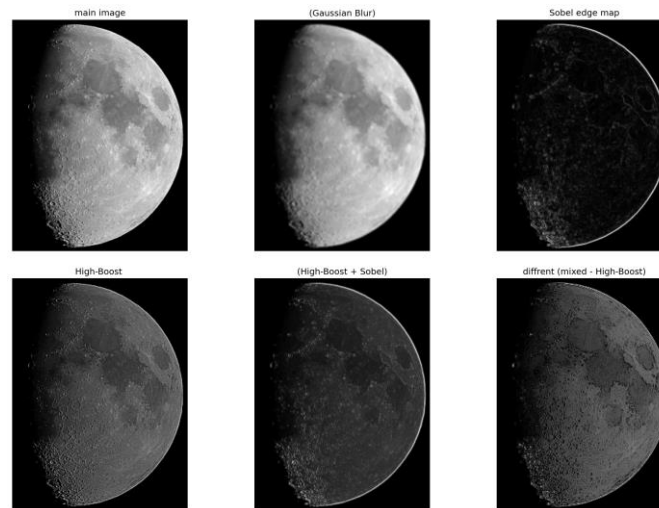


Answers:



Question 1:

Why does combining High-Boost filter with Sobel usually produce better results than using High-Boost alone?

Answer:

I think that High-Boost alone amplifies all high-frequency components uniformly, including both important edges and unwanted noise. This often leads to oversharpening. Sobel extracts only the actual edge information by calculating gradient magnitude, which represents structural boundaries in the image. When combined with Sobel, the High-Boost filter gets "guided" by the edge map, enhancing only the regions that truly contain edges. This selective enhancement preserves image structure while preventing over-amplification of smooth regions. The Sobel map acts as a spatial weighting function that tells the filter "where to sharpen" rather than sharpening everywhere equally.

Question 2:

How does changing the alpha parameter affect the final output?

Answer:

- **Alpha = 0:** Output is pure High-Boost filtering. Results in maximum. Good for images with strong edges and low noise.

- **Alpha = 1:** Output is pure Sobel edge map. Shows only edge magnitudes, not the original image content. Useful only for edge detection, not for image enhancement.
- **Alpha between 0.3 - 0.7:** Optimal range where edges are enhanced while maintaining image content. Sobel map prevents excessive sharpening in flat regions.
- **Very high alpha (>0.8):** The output becomes dominated by edge information, making the image look like an edge map overlaid on the original. Dark regions become too dark, and smooth areas lose their natural appearance.
- **Very low alpha (<0.2):** High-Boost dominates, causing noise amplification and potential oversharpening artifacts.

Question 3:

From a frequency domain perspective, what information does Sobel extract? What is the probable reason for oversharpening or halo artifacts?

Answer:

Frequency domain perspective:

Sobel operator acts as a high-pass filter that extracts:

- Vertical edges (high frequencies in horizontal direction)
- Horizontal edges (high frequencies in vertical direction)
- Diagonal edges (combination of both)

In frequency domain, Sobel emphasizes mid and high-frequency components while attenuating low frequencies (smooth regions). The gradient magnitude represents the energy at these frequencies.

Reason for oversharpening and halos:

Halos occur when high-frequency components are amplified too aggressively near sharp transitions. Mathematically, when $A > 1$ in High-Boost, the filter adds back more than the blurred component removed. At strong edges, this creates under-shoots (dark halos) on the dark side and over-shoots (bright halos) on the bright side.

Question 4:

If the image has Gaussian noise, what problem does this hybrid method create? Suggest a practical solution.

Answer:

Problem created:

Gaussian noise contains high-frequency components that appear as false edges in both the Sobel map and High-Boost filter. The Sobel operator, being sensitive to intensity changes, amplifies noise into many small "edges" that don't represent real structures. When combined with High-Boost, these false edges get enhanced, resulting in a noisy, grainy output with amplified noise patterns. The gradient calculation ($\text{sobel_x}^2 + \text{sobel_y}^2$) squares the noise, making it even more prominent.

Solutions: Pre-filtering with larger Gaussian kernel before Sobel computation, apply a Gaussian blur with larger kernel (e.g., $\text{ksize}=5$ or 7) to suppress high-frequency noise.